Global UWB System: A High-Accuracy Mobile Robot Localization System With Tightly Coupled Integration

Sy-Hung Bach, Phan-Bui Khoi, and Soo-Yeong Yi

***Abstract*—Autonomous mobile robots (AMRs) play a crucial role in smart factories and Internet of Things (IoT) applications. This article presents a high-accuracy localization system called the global UWB system (GUS) for the AMR based on ultra-wideband (UWB) distance measurements. By using two UWB modules (tags) on AMR, it is possible to estimate both position and heading angle simultaneously, which can solve the problem of error accumulation over time in internal sensors. A new tightly coupled algorithm called baseline constraint extended Kalman filter (BC-EKF) is proposed. The baseline is the actual distance between the two tags that is used as a constraint for the measurement process. The relationship between the baseline and the heading angle error is also discussed to ensure reasonable deployment of the tags. The proposed localization method is persistent as value of the geometric dilution of precision (GDOP) increases. This helps to secure the performance of the localization method even in tight workspaces where the number of fixed UWBs (anchors) is limited. The simulation and experiment results demonstrate the localization performance of the proposed method.**

***Index Terms*—Extended Kalman filter (EKF), global UWB system (GUS), robot localization, two-way ranging (TWR), ultra-wideband (UWB).**

I. Introduction

A. Motivation

Autonomous mobile robots (AMRs) are becoming popular in smart homes, smart factories, hospitals, restaurants, etc. The self-localization is the key issue determining the ability for autonomous navigation of the AMR [1]. Internal sensors such as inertial measurement units (IMUs) or odometry encoders have been commonly used in dead-reckoning localization [2], [3]. However, the dead-reckoning localization needs to be corrected periodically by external information about the environment because of the accumulation error with time [4]. External information can be detected via external sensors, such as the light detection and ranging (LiDAR) sensors [5], [6], [7], ultrasonic sensors [8], [9], [10], radio frequency identification (RFID) sensors [11], [12], [13], camera sensors [14], [15], [16], or the visible light positioning system [17]. However, these external sensors all have inherent disadvantages. Concerning LiDAR, mirror reflections and its high costs are the biggest drawback [18], [19]. Cameras are sensitive to unstable light environments [20], [21]. The ultrasonic sensor has limitation in the signal strength and beam width [22], [23]. In the context of RFID, two commonly employed localization schemes are phase-based methods and received signal strength-based methods. The principal drawback associated with received signal strength-based methods is their susceptibility to multipath effects, leading to unsatisfactory localization accuracy [24]. On the other hand, phase-based methods often entail higher hardware costs, as they typically necessitate the use of more expensive RFID readers, such as the Impinj R420 [25].

In the past few years, ultra-wideband (UWB) technology has gained significant attention from researchers for its potential in indoor localization systems [26]. Various research has been carried out to address different aspects of this technology including channel modeling [27], antenna delay calibration [28], multipath component estimation [29], orientation-dependent error [30], and scalability [31]. Furthermore, UWB sensors have emerged as an ideal choice for indoor localization with advantages, such as minimizing the multipath phenomena, high accuracy, a long measuring distance, and low cost [32], [33], [34]. Therefore, this study focuses on using the UWB for the self-localization of the AMR.

B. Related Work

In [35], an AMR positioning method based on the least-squares (LSs) algorithm using only UWB measurements was implemented in an open environment. However, the ranging accuracy was deteriorated by the high-frequency range of the UWB when it was deployed in an indoor environment with numerous opaque objects. Because the UWB alone cannot ensure the accuracy and stability of the localization, a localization method combining the internal sensors and UWB sensors was proposed based on the extended Kalman filter (EKF) [36], [37], [38]. The above studies used a loosely coupled integration approach, resulting in degraded localization performance [39]. On the other hand, a tightly

Manuscript received 4 April 2023; revised 26 September 2023; accepted 10 January 2024. Date of publication 16 January 2024; date of current version 25 April 2024. This work was supported by the Seoul National University of Science and Technology. *(Corresponding author: Soo-Yeong Yi.)*

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Digital Object Identifier 10.1109/JIOT.2024.3354786

TABLE I
RELATED WORK OVERVIEW

Ref.	Algorithm	Time synchronization non-requirement	Position correction	Heading angle correction
[35]	LS		✓	
[36]	Loosely-EKF		✓	
[37]	Loosely-EKF	✓	✓	
[38]	Loosely-EKF	✓	✓	
[40]	Tightly-EKF		✓	
[41]	Tightly-EKF		✓	
[42]	ESKF		✓	
[43]	ESKF	✓	✓	
[44]	ESKF		✓	
[45]	PF	✓	✓	
[46]	PF	✓	✓	
This work	BC-EKF	✓	✓	✓

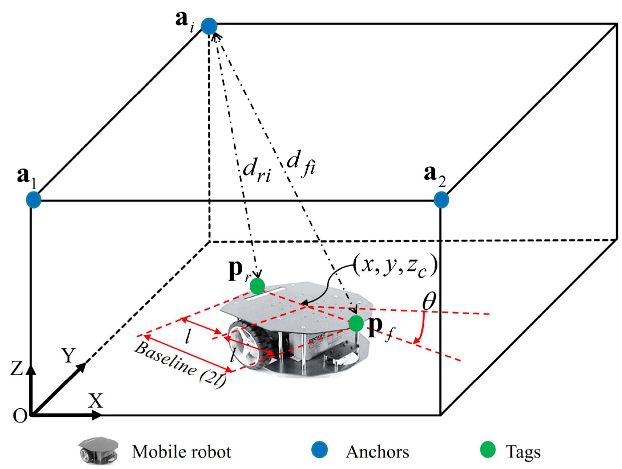

Fig. 1. Overall structure of the GUS.

coupled integration was implemented for combining the UWB measurement and the odometry data of AMR in [40] and [41], where the UWB measurement methods require time synchronization between UWB modules to ensure measurement accuracy, thereby increasing the cost and complexity of the localization system. The main difference between the two integration approaches is how the UWB measurements are used for the correction step of the EKF algorithm. In a tightly coupled integration, UWB measurements are fed directly into the EKF algorithm. In contrast, a loosely coupled integration is characterized by the indirect usage of the UWB measurements; the UWB measurements are used to calculate the AMR's pose through the trilateration method and this pose is then fed into the EKF algorithm.

A variant of the Kalman filter called the error state Kalman filter (ESKF) for inertial sensors was proposed to combine IMU and UWB data [42], [43], [44]. A particle filter (PF) localization method was also proposed to address the non-Gaussian distribution of the UWB measurement error under the non-line-of-sight (NLOS) conditions [45], [46]. However, the computational complexity of this algorithm increases with the number of particles, making it difficult to balance the computational burden and convergence time. A summary of the related works is presented in Table I.

C. Proposed System and Contributions

In the previous researches, the UWB sensors have been utilized only to estimate the AMR's position and the estimation of the heading angle was relied entirely on internal sensors. In this case, the heading angle error can be increased over time due to inadequate correction. On the contrary, two UWB tags are adopted on the AMR in the proposed localization system in order to estimate not only the position but also the heading angle of the AMR as shown in Fig. 1. Furthermore, a new tightly coupled algorithm called baseline constraint EKF (BC-EKF) is addressed for the self-localization of the AMR. Herein, the baseline is the actual distance between the two tags on the AMR. The effect of the baseline on the heading angle error is discussed to deploy the UWB tags properly. In the field of the localization based on distance measurements, the geometric dilution of precision (GDOP) is an important parameter to evaluate the localization performance [47]. In general, the localization error increases according to the GDOP value [47]. It was demonstrated in previous studies that increasing the number of anchors in the environment can help to reduce the GDOP value and improve the localization accuracy [48], [49]. However, increasing the number of anchors demands complexity and cost of the localization system. The proposed method also shows the persistent localization performance with the fixed number of anchors as the GDOP value increases. This feature makes the proposed method more effective in indoor environments.

The main contributions of this article are summarized as follows.

1) This article presents a new localization system for the AMR named the global UWB system (GUS) that can correct both position and heading angle by UWB measurements.
2) A new tightly coupled algorithm called BC-EKF is proposed to combine the odometry encoder and UWB data, where the baseline is used as a constraint in the correction step.
3) Effects of the baseline on the estimation of the heading angle has been discussed for proper deployment of the tags on the AMR.
4) The localization performance of the proposed method is persistent against the GDOP value than that of the existing methods.

The remainder of this article is organized as follows. Section II discusses the overall structure of the GUS and the problems of the UWB measurement. Section III proposes the framework and the mathematical model of the UWB and encoder data fusion algorithm. In Section IV, the localization performance of different algorithms is compared and discussed through simulation and experimental results. In Section V, concluding remarks are presented.

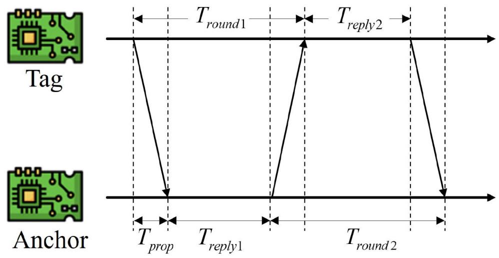

Fig. 2. SDS-TWR principle for UWB distance measurement.

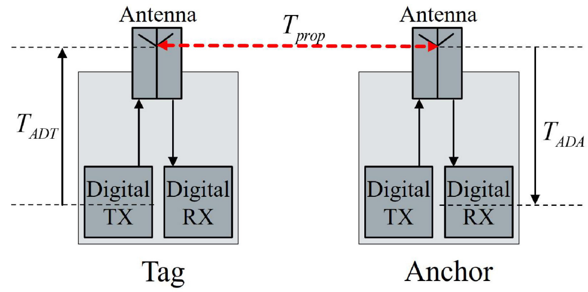

Fig. 3. Antenna delay diagram.

II. GUS

In this section, the overall structure of the proposed localization system is described. Because the accuracy of distance measurements is affected by environmental conditions, the calibration and measurement methods for the UWB modules are presented also.

A. Structure

The overall structure of the GUS is presented in Fig. 1. In the global coordinate system, anchors are fixed at known positions, $\mathbf{a}_i = [x_{ai}\ y_{ai}\ z_{ai}]^T, i = 1, \ldots, 3$ in the environment. In this study, it is assumed that the AMR moves in 2-D plane, and two tags mounted on the AMR have the same height with constant z_c. Then, the position vectors of the front and the rear tags are represented as $\mathbf{p}_f = [x_f\ y_f\ z_c]^T$ and $\mathbf{p}_r = [x_r\ y_r\ z_c]^T$, respectively. The measured distances from the anchor $\mathbf{a}_i$ to the front and the rear tags are denoted as d_{fi} andd_{ri}. The pose vector of the AMR is represented as $\mathbf{x} = [x\ y\ \theta]^T$, where θ is the heading angle. The distance between the tags and the AMR's center is denoted as l in Fig. 1 and the baseline distance is $2l$ in this case.

B. UWB Ranging

Among the several methods of distance measurement between an anchor and a tag in UWB technology, a time-of-flight (TOF) method named symmetric-double-sided two-way ranging (SDS-TWR) is generally used because it has high accuracy without time synchronization. By using timestamps on each UWB module as shown in Fig. 2, the intervals T_{round1}, T_{reply1}, T_{round2}, and T_{reply2}, are obtained and the TOF, T_{prop}, from a tag to an anchor can be calculated as follows [50]:

$$T_{\text{prop}} = \frac{T_{\text{round1}} \times T_{\text{round2}} - T_{\text{reply1}} \times T_{\text{reply2}}}{T_{\text{round1}} + T_{\text{reply1}} + T_{\text{round2}} + T_{\text{reply2}}}. \quad (1)$$

Then, the distance can be computed by using T_{prop} and the propagation speed, c, of the electromagnetic waves as follows:

$$d = cT_{\text{prop}}. \quad (2)$$

The distance measurement rate of SDS-TWR is typically high, with up to around one hundred measurements per second for a pair of UWB modules. The high-rate distance measurement of SDS-TWR makes it suitable for the indoor localization system [51].

C. UWB Calibration

UWB range can achieve centimeter-level accuracy and a maximum measurable range of more than 100 m in outdoor environments. However, both of them have been significantly degraded when UWB is operated in indoor environments [52]. Therefore, the preliminary calibration is necessary to use UWB for indoor environments. The UWB calibration procedure for distance measurements using the SDS-TWR method is divided into two steps: 1) antenna delay calibration and 2) bias calibration.

1) *Antenna Delay Calibration:* When measuring the TOF, the measured time, T_{measured}, includes the antenna delay as shown in Fig. 3

$$T_{\text{measured}} = T_{\text{ADT}} + T_{\text{prop}} + T_{\text{ADA}} \quad (3)$$

where T_{ADT} and T_{ADA} denote the antenna delays of tag and anchor and T_{prop} represents the pure TOF between a tag and an anchor. According to (3), the sum of T_{ADT} and T_{ADA} must be corrected to ensure that T_{prop} is exactly obtained. By setting up a tag and an anchor with known distances and setting T_{ADT} equal to the average delay provided by the manufacturers, a trial-and-error loop to find the T_{ADA} has been proposed in [53]. In this study, the following process is proposed to calibrate the antenna delay for two tags and three anchors in the GUS structure.

   a) *Step 1:* Set the T_{ADT} of Tag-1 equal to the average delay value provided by the manufacturers.
   b) *Step 2:* Use Tag-1 to find the value T_{ADA} of the three anchors in turn.
   c) *Step 3:* Use Anchor-1 with the T_{ADA} found in step 2 to determine the T_{ADT} of Tag-2.

2) *Bias Calibration:* Antenna delay calibration only guarantees accuracy for a specific distance. Therefore, bias error calibration is necessary to ensure that UWB can accurately measure the various distances within the coverage area.

It is assumed that the relationship between the measured distance, d, and the actual distance, $\bar{d}$, is expressed as follows:

$$\bar{d} = (1 + \alpha)\ d + \beta \quad (4)$$

where α and β denotes the paraments of the bias calibration which can be obtained by taking multiple measurements at different distances [54].

III. UWB and Odometry Data Fusion Algorithm

In this section, a new tightly coupled algorithm called BC-EKF is proposed for the self-localization problem. To implement the algorithm, 1) a prediction model based on odometry and 2) an observation model using the UWB distance measurement to correct the prediction are addressed. In addition, the influence of the baseline and the GDOP on the performance of the proposed algorithm is also discussed.

A. Prediction Model With Odometry

The AMR's state equation in the discrete-time domain is described as follows:

$$\begin{aligned}\mathbf{x}_k &= \mathbf{f}(\mathbf{x}_{k-1}, \mathbf{u}_k, \mathbf{q}_k)\\ &= \begin{bmatrix} x_{k-1} + v_k T \cos\left(\theta_{k-1} + \frac{\omega_k T}{2}\right) + q_{x,k} \\ y_{k-1} + v_k T \sin\left(\theta_{k-1} + \frac{\omega_k T}{2}\right) + q_{y,k} \\ \theta_{k-1} + \omega_k T + q_{\theta,k} \end{bmatrix}\end{aligned} \tag{5}$$

where T is the sampling interval, $\mathbf{x}_k = [x_k \; y_k \; \theta_k]^T$represents the state vector,$\mathbf{u}_k = [v_k \; \omega_k]^T$denotes the input vector from the odometry sensor, and $\mathbf{q}_k = [q_{x,k} \; q_{y,k} \; q_{\theta,k}]^T$is the Gaussian white noise with covariance matrix $\mathbf{Q} = \text{diag}[\sigma_{q_x}^2 \; \sigma_{q_y}^2 \; \sigma_{q_\theta}^2]$.

The Jacobian matrix, $\mathbf{A}_k$, can be obtained from (5) by linearization as follows:

$$\mathbf{A}_k = \frac{\partial\, \mathbf{f}(\mathbf{x}_k, \mathbf{u}_k, 0)}{\partial\, \mathbf{x}_k} = \begin{bmatrix} 1 & 0 & -v_k T \sin\left(\theta_k + \frac{\omega_k T}{2}\right) \\ 0 & 1 & v_k T \cos\left(\theta_k + \frac{\omega_k T}{2}\right) \\ 0 & 0 & 1 \end{bmatrix}. \tag{6}$$

B. Observation Model With Baseline Constraint

The observation vector at time k can be represented by UWB distance measurements $d_{fi,k}$ and $d_{ri,k}$ as follows:

$$\begin{aligned}\mathbf{z}_k &= \mathbf{h}(\mathbf{x}_k, \mathbf{w}_k)\\ &= \begin{bmatrix} \sqrt{(x_{f,k} - x_{\mathbf{a}_1})^2 + (y_{f,k} - y_{\mathbf{a}_1})^2 + (z_c - z_{\mathbf{a}_1})^2} + w_{f1,k} \\ \vdots \\ \sqrt{(x_{f,k} - x_{\mathbf{a}_n})^2 + (y_{f,k} - y_{\mathbf{a}_n})^2 + (z_c - z_{\mathbf{a}_n})^2} + w_{fn,k} \\ \sqrt{(x_{r,k} - x_{\mathbf{a}_1})^2 + (y_{r,k} - y_{\mathbf{a}_1})^2 + (z_c - z_{\mathbf{a}_1})^2} + w_{r1,k} \\ \vdots \\ \sqrt{(x_{r,k} - x_{\mathbf{a}_n})^2 + (y_{r,k} - y_{\mathbf{a}_n})^2 + (z_c - z_{\mathbf{a}_n})^2} + w_{rn,k} \end{bmatrix}\end{aligned} \tag{7}$$

where $\mathbf{w}_k = [w_{fi,k} \; w_{ri,k}]^T$is the Gaussian white noise with covariance matrix $\mathbf{R} = \text{diag}[\sigma_{w_{fi}}^2 \sigma_{w_{ri}}^2]$. Here, the subscript $i = 1, \ldots, n$ denotes the anchor index, with n being the number of anchors. In the case of using three anchors (n=3) in the GUS, the dimension of $\mathbf{h}(.)$ is 6×1.

The baseline constraint between the two tags on the AMR is described as

$$\mathbf{p}_{f,k} = \begin{bmatrix} x_k + l\cos\theta_k \\ y_k + l\sin\theta_k \\ z_c \end{bmatrix}, \quad \mathbf{p}_{r,k} = \begin{bmatrix} x_k - l\cos\theta_k \\ y_k - l\sin\theta_k \\ z_c \end{bmatrix}. \tag{8}$$

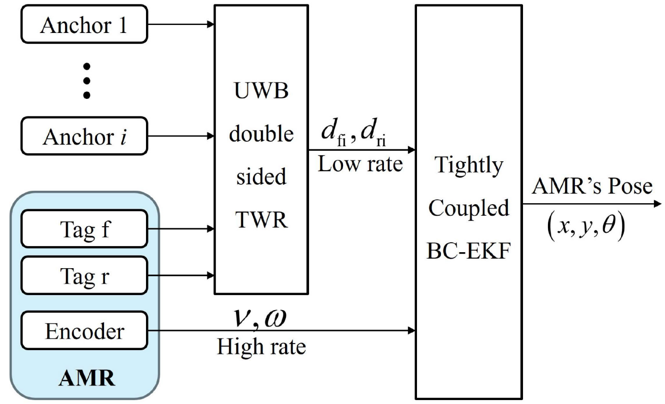

Fig. 4. Architecture of the tightly coupled BC-EKF algorithm.

From (7) and (8), the Jacobian matrix, $\mathbf{H}_k$, can be obtained as

$$\mathbf{H}_k = \frac{\partial\, \mathbf{h}(\mathbf{x}_k, 0)}{\partial\, \mathbf{x}_k} = \begin{bmatrix} \frac{x_{f,k} - x_{\mathbf{a}_1}}{D_{f1,k}} & \frac{y_{f,k} - y_{\mathbf{a}_1}}{D_{f1,k}} & \frac{C_{f1,k}}{D_{f1,k}} \\ \vdots & \vdots & \vdots \\ \frac{x_{f,k} - x_{\mathbf{a}_n}}{D_{fn,k}} & \frac{y_{f,k} - y_{\mathbf{a}_n}}{D_{fn,k}} & \frac{C_{fn,k}}{D_{fn,k}} \\ \frac{x_{r,k} - x_{\mathbf{a}_1}}{D_{r1,k}} & \frac{y_{r,k} - y_{\mathbf{a}_1}}{D_{r1,k}} & \frac{C_{r1,k}}{D_{r1,k}} \\ \vdots & \vdots & \vdots \\ \frac{x_{r,k} - x_{\mathbf{a}_n}}{D_{rn,k}} & \frac{y_{r,k} - y_{\mathbf{a}_n}}{D_{rn,k}} & \frac{C_{rn,k}}{D_{rn,k}} \end{bmatrix} \tag{9}$$

where $C_{fi,k}$, $C_{ri,k}$, $D_{fi,k}$, and $D_{ri,k}$ are given as follows:

$$\begin{aligned} C_{fi,k} &= -(x_{f,k} - x_{\mathbf{a}_i})l\sin\theta_k + (y_{f,k} - y_{\mathbf{a}_i})l\cos\theta_k \\ C_{ri,k} &= (x_{r,k} - x_{\mathbf{a}_i})l\sin\theta_k - (y_{r,k} - y_{\mathbf{a}_i})l\cos\theta_k \\ D_{fi,k} &= \sqrt{(x_{f,k} - x_{\mathbf{a}_i})^2 + (y_{f,k} - y_{\mathbf{a}_i})^2 + (z_c - z_{\mathbf{a}_i})^2} \\ D_{ri,k} &= \sqrt{(x_{r,k} - x_{\mathbf{a}_i})^2 + (y_{r,k} - y_{\mathbf{a}_i})^2 + (z_c - z_{\mathbf{a}_i})^2}. \end{aligned} \tag{10}$$

C. Tightly Coupled BC-EKF Algorithm

The architecture of the tightly coupled BC-EKF algorithm is shown in Fig. 4 where six UWB measurements are fed directly into the BC-EKF algorithm. The BC-EKF is operated at a high rate of odometry to predict the AMR's state. A correction will be performed whenever the UWB measurement is feasible. The BC-EKF algorithm is described as follows:

1) *Prediction:* A priori estimation $\widehat{\mathbf{x}}_k^-$ for $\mathbf{x}_{k-1}$ is given by (5)

$$\widehat{\mathbf{x}}_k^- = \mathbf{f}(\mathbf{x}_{k-1}, \mathbf{u}_k, 0) \tag{11}$$

$$\widehat{\mathbf{P}}_k^- = \mathbf{A}_k \mathbf{P}_{k-1} \mathbf{A}_k^T + \mathbf{Q}. \tag{12}$$

2) *Correction:* The priori estimation, $\widehat{\mathbf{x}}_k^-$, is corrected by UWB measurements resulting in a posteriori estimation, $\mathbf{x}_k$, as follows:

$$\mathbf{K}_k = \widehat{\mathbf{P}}_k^- \mathbf{H}_k^T \left(\mathbf{H}_k \widehat{\mathbf{P}}_k^- \mathbf{H}_k^T + \mathbf{R}\right)^{-1} \tag{13}$$

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k)\widehat{\mathbf{P}}_k^- \tag{14}$$

$$\mathbf{x}_k = \widehat{\mathbf{x}}_k^- + \mathbf{K}_k\left\{\mathbf{z}_k - \mathbf{h}\left(\widehat{\mathbf{x}}_k^-, 0\right)\right\}. \tag{15}$$

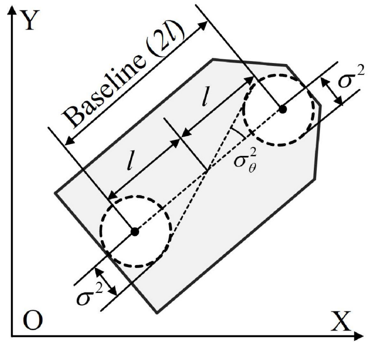

Fig. 5. Effect of baseline to heading angle error.

D. Effect of Baseline on Heading Angle Error

The heading angle of the AMR can be estimated through the estimated positions of the two tags. It is assumed that the estimation error variance of each tag position is equal as σ^2. Then, the estimation error variance of the heading angle shown in Fig. 5 can be obtained as follows:

$$\sigma_\theta^2 = \tan^{-1}\left(\frac{\sigma^2}{l}\right). \tag{16}$$

From (16), it is possible to reduce the estimation error of the heading angle by increasing the baseline distance. However, in actual application, the maximum allowable baseline depends on the AMR's size. It is necessary to investigate the influence of the baseline on the algorithms, which will be presented in the next section.

E. GDOP Model

The GDOP index has been proposed to evaluate the influence of the geometric distribution of anchors on the performance of tag localization [47]. In general, the higher GDOP is, the larger the positioning error is. The GDOP can be computed by the following formula [48]:

$$\text{GDOP} = \sqrt{\text{tr}\left(\mathbf{B}^T\mathbf{B}\right)^{-1}} \tag{17}$$

where $\mathbf{B}$ is given at an AMR pose,$\mathbf{x}_k = [x_k \; y_k \; \theta_k]^T$ as follows:

$$\mathbf{B} = \begin{bmatrix} \frac{x_{\mathbf{a}_1} - x_k}{\sqrt{(x_{\mathbf{a}_1}-x_k)^2 + (y_{\mathbf{a}_1}-y_k)^2}} & \frac{y_{\mathbf{a}_1} - y_k}{\sqrt{(x_{\mathbf{a}_1}-x_k)^2 + (y_{\mathbf{a}_1}-y_k)^2}} \\ \vdots & \vdots \\ \frac{x_{\mathbf{a}_n} - x_k}{\sqrt{(x_{\mathbf{a}_n}-x_k)^2 + (y_{\mathbf{a}_n}-y_k)^2}} & \frac{y_{\mathbf{a}_n} - y_k}{\sqrt{(x_{\mathbf{a}_n}-x_k)^2 + (y_{\mathbf{a}_n}-y_k)^2}} \end{bmatrix}. \tag{18}$$

It is possible to obtain a lower GDOP in narrow environments by increasing the number of anchors [48], [49]. However, a higher number of anchors demands the cost and complexity of the localization system [55]. The effect of the GDOP on the localization performance of the proposed BC-EKF algorithm in this study is evaluated with simulations and experiments.

IV. Simulation and Experimental Results

Three localization methods are selected to compare and verify the proposed method.

1) *BC-EKF:* The BC-EKF proposed in this article is implemented with two tags in the GUS. The baseline distance of the two tags is used as a constraint for the EKF correction step.
2) *1-Tag-EKF:* One UWB tag and odometry data are used based on the tightly coupled EKF method [40], [41].
3) *Trilateration:* The position in 2-D space is directly estimated from the distances between a tag and three anchors [56]. The AMR's pose is then determined through the estimated positions of the two tags.

A. Simulations

1) Accuracy of Localization Methods: In this simulation, it is assumed that 1) an AMR moves on the x–y plane with an initial pose, $\mathbf{x}_0 = [2.5 \; 0 \; 0]^T$; 2) the linear and angular velocities are $\nu = 0.3$ m/s and $\omega = 7.5$ deg/s, respectively; 3) the total travel time is $N = 50$ s with the sample time period $T = 0.05$ s; 4) the update frequencies of the odometry for the prediction and the UWB measurement for the correction are $f_{\text{pred}} = 20$ Hz and $f_{\text{corr}} = 5$ Hz, respectively; and 5) the coordinates of the anchors are $\mathbf{a}_1 = [0 \; 0 \; 1]^T$, $\mathbf{a}_2 = [0 \; 5 \; 1]^T$, and $\mathbf{a}_3 = [5 \; 0 \; 1]^T$.

The simulation is performed with different sets of $\mathbf{Q}$ and $\mathbf{R}$ values, where $\mathbf{Q}$ and $\mathbf{R}$ represent the process and observation noise variances, respectively. By fixing $\mathbf{Q}$ and varying $\mathbf{R}$ with different values for each simulation, the influence of $\mathbf{R}$ on the performance of the three methods is summarized as follows.

1) *Trajectory:* As shown in Fig. 6(a)–(c), oscillations in the trajectories of all three methods decrease when the value of $\mathbf{R}$ becomes smaller, indicating that $\mathbf{R}$ affects all of them. However, the trilateration method is more sensitive to the change of $\mathbf{R}$.
2) *Position Error:* Fig. 6(a-1)–(c-1) shows the cumulative distribution function (CDF) plots of position error, where the performance of the trilateration method is more sensitive with the value $\mathbf{R}$ than those of the other two methods. The difference in position error of the 1-Tag-EKF and BC-EKF methods is almost unchanged when $\mathbf{R}$ changes, which shows that the effect of $\mathbf{R}$ on position error is equivalent for the two methods.
3) *Heading Angle Error:* The heading angle error of the trilateration method decreases with decreasing the $\mathbf{R}$ value. Although the mean heading angle error is reduced with decreasing $\mathbf{R}$ for 1-Tag-EKF, the maximum heading angle error is almost unchanged as shown in Fig. 6(a-2)–(c-2).

In summary, the simulation results demonstrate that the BC-EKF proposed in this article outperforms the other two methods in terms of both position and heading angle performances. The BC-EKF has resolved the $\mathbf{R}$-sensitivity problem of the trilateration method. Particularly noteworthy is the significant reduction in the maximum heading angle error when employing the proposed method, a considerable improvement compared to the 1-Tag-EKF. The 1-Tag-EKF shows a substantial maximum heading angle error, which remains constant as $\mathbf{R}$ is reduced, signifying its inadequacy in

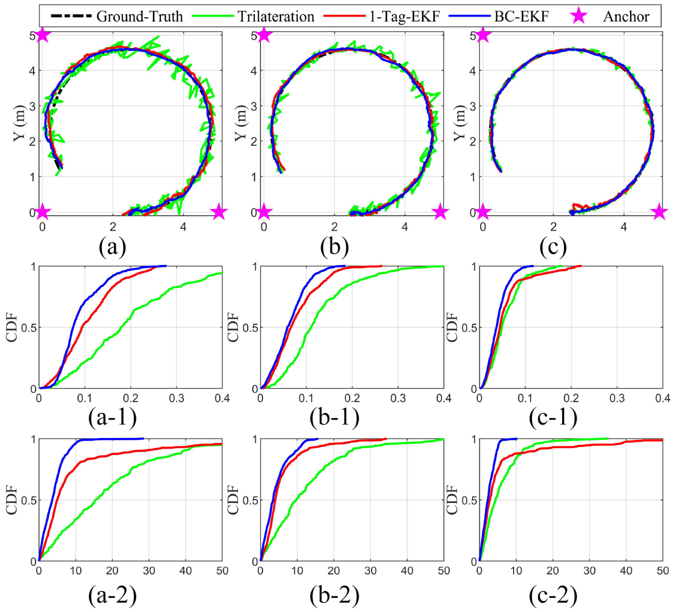

Fig. 6. Localization performances when $\mathbf{Q} = 0.0001$ with different $\mathbf{R}$ values: (a), (a-1), and (a-2) represent trajectory, CDF of position and heading angle error with $\mathbf{R} = 0.0225$; (b), (b-1), and (b-2) represent trajectory, position, and heading angle error CDF with $\mathbf{R} = 0.01$; and (c), (c-1), and (c-2) represent trajectory, position, and heading angle error CDF with $\mathbf{R} = 0.0025$.

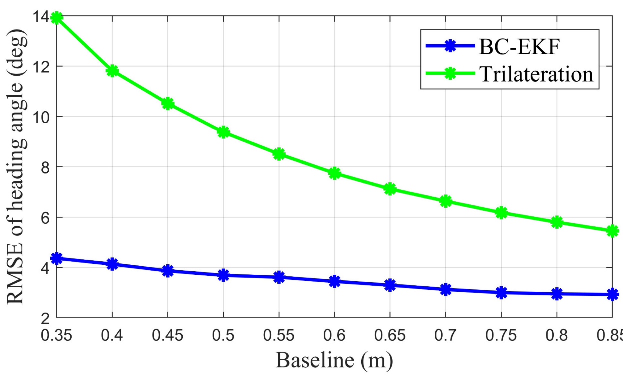

Fig. 7. Simulation results: effect of baseline on heading angle error.

fully correcting heading angles. On the other hand, the BC-EKF method results have the decreased maximum heading angle error as $\mathbf{R}$ is decreased.

2) Effect of Baseline Distance Between Two Tags on Heading Angle Error: In order to evaluate the effect of the baseline on heading angle error, simulations are performed with baseline values ranging from 0.35 to 0.85 m with $\mathbf{Q} = 0.0001$ and $\mathbf{R} = 0.0025$. The root-mean-square error (RSME) of the heading angle error is calculated over the entire simulation trajectory with each baseline distance and is represented in Fig. 7. The simulation results show that the RSME of the heading angle error decreases as shown in (16) as the baseline distance of the two tags on the AMR increases. The BC-EKF method proposed in this study is always superior to the trilateration method in terms of heading angle estimation accuracy, especially for the cases where the baseline is short. This allows the BC-EKF approach to secure the autonomous

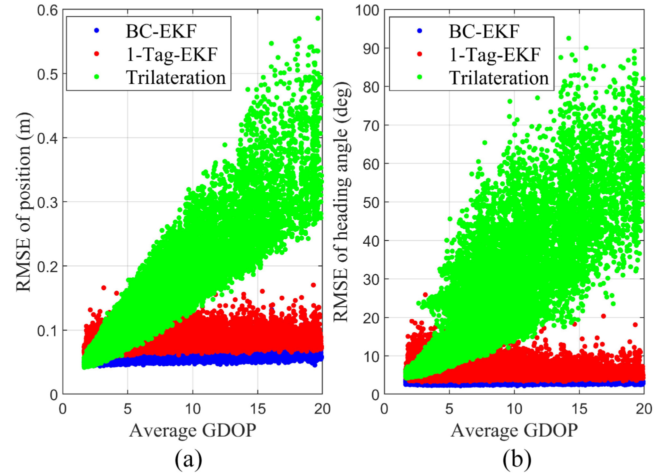

Fig. 8. Simulation results: effect of GDOP on the localization performance. (a) RMSE of position. (b) RMSE of heading angle.

navigation of the AMR even when it is moving in a narrow workspace, where the robot size and the baseline are limited.

3) Effect of GDOP on Localization Methods: In this simulation, the 5×5 m^2 working area is divided into grids with 0.5×0.5 m. Three anchors are relocated to all different mesh points within the grid with noncollinear and nonoverlapping conditions. The following parameters are calculated over the entire simulation trajectory for each anchor configuration: 1) RSME of position and heading angle and 2) average GDOP by (17) and (18). Fig. 8 shows the effect of the GDOP on the performance of the different localization methods when $\mathbf{Q}$, $\mathbf{R}$, and the baseline are 0.0001, 0.0025, and 0.65 m, respectively. In Fig. 8(a) and (b), each dot implies the RSME of position or heading angle on the overall trajectory for each corresponding anchor configuration with the average GDOP. The localization error of the trilateration method increases according to the GDOP; it is difficult to apply this method in tight environments. In contrast, the localization performance of the 1-Tag-EKF and the BC-EKF is less influenced by the GDOP, and they have accurate localization performances even in tight workspaces with a relatively small number of anchors.

B. Experiment

1) Setup: An experimental setup is shown in Fig. 9 where the following components are used: 1) a high-accuracy LiDAR (Hokuyo ust-10lx) to extract the ground-truth data; 2) three anchors and two tags made up of DWM1000 modules and ESP32 microcontrollers; and 3) an AMR prototype.

By applying the UWB calibration procedure presented in Section II, the UWB measurement can achieve an accuracy of 0.1 m in indoor environments. The calibration results are shown in Fig. 10.

2) Accuracy of Localization Methods: Experimental results shown in Fig. 11 are obtained with the following parameters: 1) the positions of the three anchors are $\mathbf{a}_1 = [-3\ -1.2\ 1]^T$, $\mathbf{a}_2 = [3\ -1.2\ 1]^T$, and $\mathbf{a}_3 = [0\ 3.8\ 1]^T$; 2) the baseline is 0.65 m; 3) process noise variance $\mathbf{Q} = 0.0001$; 4) the measurement noise variance $\mathbf{R} = 0.0016$; 5) the total travel

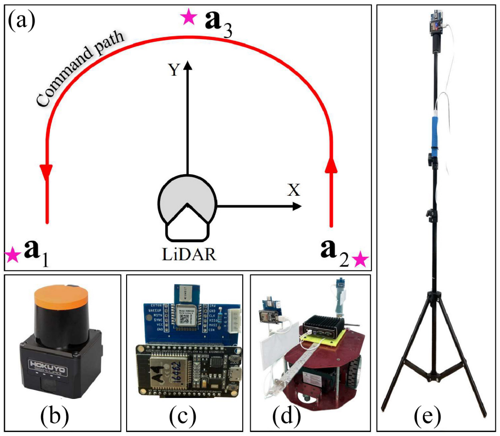

Fig. 9. Experimental setup: (a) structure of experiment, (b) LiDAR for extract ground-truth data, (c) UWB and microprocessor module, (d) AMR, and (e) anchor on tripod.

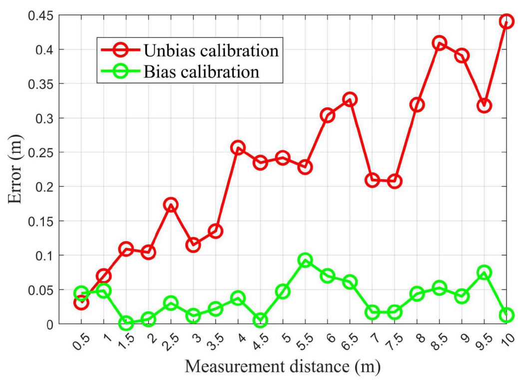

Fig. 10. Experimental results of UWB calibration.

time $N = 56$ s and the sampling time period $T = 0.05$ s; and 6) the odometry update frequency is 20 Hz and the UWB measurement frequency is 5 Hz. As shown in Fig. 11(a), the trajectory estimated by the BC-EKF is less volatile and more closely follows the ground-truth trajectory than the other two methods. The CDF error of the position and the heading angle are shown in Fig. 11(b) and (c), respectively. The quantitative comparison of the localization performance is summarized in Table II.

3) Effect of Baseline to Heading Angle Error: In order to verify the effect of the baseline on the heading angle estimation, experiments are carried out. The RSME of the heading angle estimation on trajectory is calculated for each baseline distance. Fig. 12 shows the results of experiments that the RSME of the heading angle estimation by the proposed BC-EKF method is approximately half that of the trilateration method. This experimental result can be used for AMR design considering the achievable accuracy of the heading angle estimation.

4) Effect of GDOP on Localization Methods: Nine anchor configurations with increasing GDOP values from 1.43 to 18.05 are designed and tested as shown in Table III. The experimental results in Fig. 13 show that the proposed BC-EKF method is less influenced by GDOP, which can be expected by the simulation results in Fig. 8. This makes the BC-EKF method efficiently applicable in indoor environments where the working space is narrow such as building corridors or small rooms.

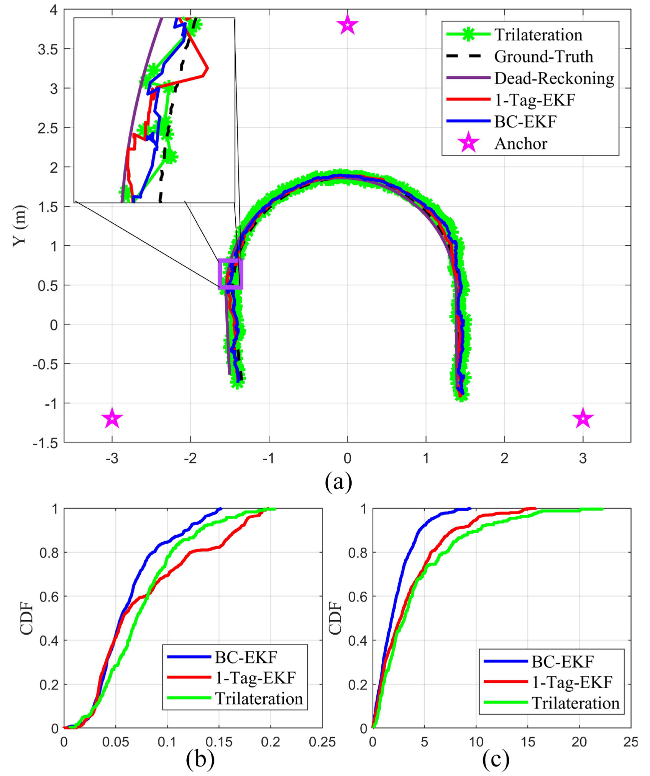

Fig. 11. Experimental results: comparison of the localization accuracy. (a) X (m). (b) Position error (m). (c) Heading angle error (deg).

TABLE II
COMPARISON OF LOCALIZATION ACCURACY

Method	Position error (m)		Heading angle error (deg)	
	Max	RSME	Max	RSME
Trilateration	0.2040	0.0860	22.2019	5.9444
1-Tag-EKF	0.1982	0.0944	15.7755	4.7766
BC-EKF	**0.1523**	**0.0714**	**9.4489**	**2.8597**

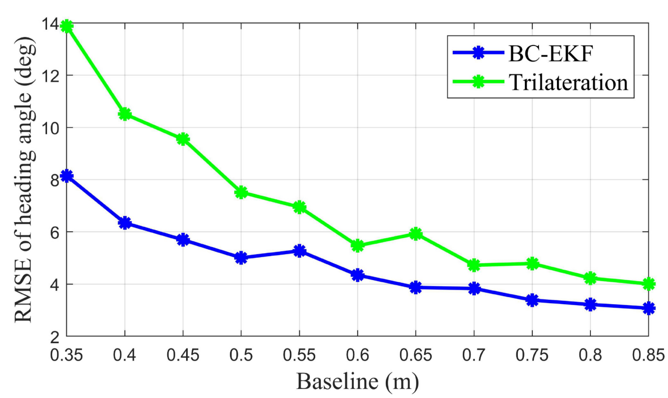

Fig. 12. Experimental results: effect of baseline on heading angle error.

TABLE III
NINE ANCHOR CONFIGURATIONS WITH INCREASING GDOP

No.	$\mathbf{a}_1$	$\mathbf{a}_2$	$\mathbf{a}_3$	average GDOP
1	$[-3\ \ -1.2\ \ 1]^T$	$[0\ \ 3.8\ \ 1]^T$	$[3\ \ -1.2\ \ 1]^T$	1.432
2	$[-1\ \ -1.2\ \ 1]^T$	$[-2\ \ -2.8\ \ 1]^T$	$[2\ \ -1.2\ \ 1]^T$	2.181
3	$[-1\ \ -1.2\ \ 1]^T$	$[-2\ \ -2.8\ \ 1]^T$	$[1\ \ -1.2\ \ 1]^T$	3.285
4	$[-0.6\ \ -0.8\ \ 1]^T$	$[-2\ \ -2.8\ \ 1]^T$	$[1\ \ -1.2\ \ 1]^T$	5.092
5	$[-0.6\ \ -0.8\ \ 1]^T$	$[-2\ \ -2.8\ \ 1]^T$	$[0.8\ \ -1.2\ \ 1]^T$	7.489
6	$[-0.4\ \ -0.8\ \ 1]^T$	$[-2\ \ -2.8\ \ 1]^T$	$[1\ \ -1.2\ \ 1]^T$	10.785
7	$[-0.4\ \ -0.6\ \ 1]^T$	$[-2\ \ -2.8\ \ 1]^T$	$[0.8\ \ -1.2\ \ 1]^T$	12.424
8	$[-0.4\ \ -0.6\ \ 1]^T$	$[-2\ \ -2.8\ \ 1]^T$	$[0.6\ \ -1.2\ \ 1]^T$	15.132
9	$[0\ \ -0.4\ \ 1]^T$	$[-2\ \ -2.8\ \ 1]^T$	$[1\ \ -1.2\ \ 1]^T$	18.051

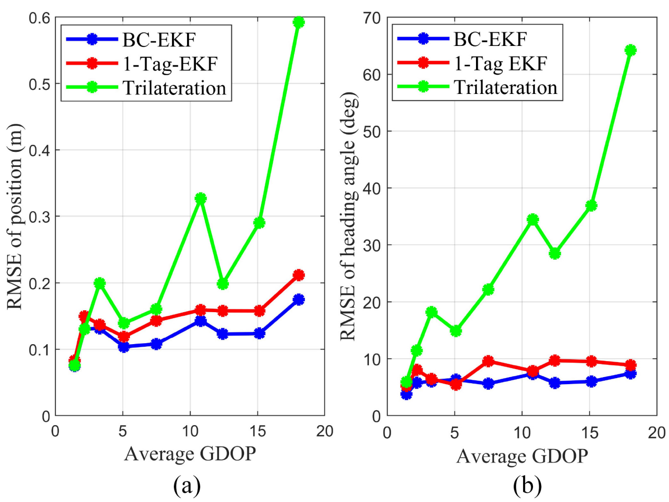

Fig. 13. Experimental results: effect of GDOP on the localization performance. (a) RMSE of position. (b) RMSE of heading angle.

V. CONCLUSION

This article presents a high-accuracy localization system called GUS that is capable of simultaneous estimation for both the position and heading angle by using two tags on AMR. The odometry data is used for the prediction of the AMR's pose with high rate. The UWB measurement is used for the correction step of EKF with relatively lower rate. In addition, an algorithm named the BC-EKF has been proposed that uses the baseline of the two tags as a constraint for the measurement process.

Simulation and experimental results have demonstrated that the proposed method has the following advantages: 1) the BC-EKF achieves better estimation accuracy than the conventional methods for both position and heading angle; 2) the heading angle estimation error is always smaller than that of the trilateration method regardless of the baseline value; and 3) the localization performance of the proposed method is less influenced by GDOP than the conventional methods. The above advantages make the proposed GUS system with the BC-EKF algorithm applicable for practical situations.

REFERENCES

[1] R. Siegwart, I. Nourbakhsh, and D. Scaramuzza, *Introduction to Autonomous Mobile Robots*, 2nd ed., Cambridge, MD, USA: MIT Press, 2011.

[2] B.-S. Cho, W.-S. Moon, W.-J. Seo, and K.-R. Baek, "A dead reckoning localization system for mobile robots using inertial sensors and wheel revolution encoding," *J. Mech. Sci. Technol.*, vol. 25, no. 11, pp. 2907–2917, 2011.

[3] B. Barshan and H. F. Durrant-Whyte, "Inertial navigation systems for mobile robots," *IEEE Trans. Robot. Autom.*, vol. 11, no. 3, pp. 328–342, Jun. 1995.

[4] A. Kelly, "Linearized error propagation in odometry," *Int. J. Robot. Res.*, vol. 23, no. 2, pp. 179–218, 2004.

[5] P. Jiang, L. Chen, H. Guo, M. Yu, and J. Xiong, "Novel indoor positioning algorithm based on Lidar/inertial measurement unit integrated system," *Int. J. Adv. Robot. Syst.*, vol. 18, no. 2, 2021, Art. no. 1729881421999923.

[6] H. Sobreira, A. P. Moreira, P. G. Costa, and J. Lima, "Robust mobile robot localization based on security laser scanner," in *Proc. IEEE Int. Conf. Auton. Robot Syst. Comp.*, Vila Real, Portugal, 2015, pp. 162–167.

[7] Z. Xu, S. Guo, T. Song, and L. Zeng, "Robust localization of the mobile robot driven by Lidar measurement and matching for ongoing scene," *Appl. Sci.*, vol. 10, no. 18, p. 6152, 2020.

[8] S. Y. Yi and B. W. Choi, "Autonomous navigation of indoor mobile robots using a global ultrasonic system," *Robotica*, vol. 22, no. 4, pp. 369–374, 2004.

[9] S. J. Kim and B. K. Kim, "Dynamic ultrasonic hybrid localization system for indoor mobile robots," *IEEE Trans. Ind. Electron.*, vol. 60, no. 10, pp. 4562–4573, Oct. 2013.

[10] C.-C. Tsai, "A localization system of a mobile robot by fusing dead-reckoning and ultrasonic measurements," *IEEE Trans. Instrum. Meas.*, vol. 47, no. 5, pp. 1399–1404, Oct. 1998.

[11] W. Gueaieb and M. S. Miah, "An intelligent mobile robot navigation technique using RFID technology," *IEEE Trans. Instrum. Meas.*, vol. 57, no. 9, pp. 1908–1917, Sep. 2008.

[12] S. Park and S. Hashimoto, "Autonomous mobile robot navigation using passive RFID in indoor environment," *IEEE Trans. Ind. Electron.*, vol. 56, no. 7, pp. 2366–2373, Jul. 2009.

[13] P. Yang and W. Wu, "Efficient particle filter localization algorithm in dense passive RFID tag environment," *IEEE Trans. Ind. Electron.*, vol. 61, no. 10, pp. 5641–5651, Oct. 2014.

[14] P. Nazemzadeh, D. Fontanelli, D. Macii, and L. Palopoli, "Indoor localization of mobile robots through QR code detection and dead reckoning data fusion," *IEEE/ASME Trans. Mechatronics*, vol. 22, no. 6, pp. 2588–2599, Dec. 2017.

[15] G. Panahandeh and M. Jansson, "Vision-aided inertial navigation based on ground plane feature detection," *IEEE/ASME Trans. Mechatronics*, vol. 19, no. 4, pp. 1206–1215, Aug. 2014.

[16] S.-H. Bach, P.-B. Khoi, and S.-Y. Yi, "Application of QR code for localization and navigation of indoor mobile robot," *IEEE Access*, vol. 11, pp. 28384–28390, 2023.

[17] Y. Zhuang, Q. Wang, M. Shi, P. Cao, L. Qi, and J. Yang, "Low-power centimeter-level localization for indoor mobile robots based on ensemble Kalman smoother using received signal strength," *IEEE Internet Things J.*, vol. 6, no. 4, pp. 6513–6522, Aug. 2019.

[18] C. C. Wang, C. Thorpe, S. Thrun, M. Hebert, and H. Durrant-Whyte, "Simultaneous localization mapping and moving object tracking," *Int. J. Robot. Res.*, vol. 26, no. 9, pp. 889–916, 2007.

[19] Y. Li and J. Ibanez-Guzman, "Lidar for autonomous driving: The principles, challenges, and trends for automotive lidar and perception systems," *IEEE Signal Process. Mag.*, vol. 37, no. 4, pp. 50–61, Jul. 2020.

[20] Y. Jeon, H. Kim, and S.-W. Seo, "ABCD: Attentive bilateral convolutional network for robust depth completion," *IEEE Robot. Autom. Lett.*, vol. 7, no. 1, pp. 81–87, Jan. 2022.

[21] K.-Y. Tu and C.-H. Chen, "Analysis of camera's images influenced by light variation," in *Proc. SICE Annu. Conf.*, Takamatsu, 2007, pp. 2089–2094.

[22] B. Hoyle and L. Xu, "Ultrasonic sensors," in *Process Tomography: Principles, Techniques and Applications*. Oxford, U.K.: Butterworth-Heinemann, 1995, pp. 119–149.

[23] S.-Y. Yi, "Global ultrasonic system with selective activation for autonomous navigation of an indoor mobile robot," *Robotica*, vol. 26, no. 3, pp. 277–283, 2008.

[24] Y. Ma, L. Zhou, K. Liu, and J. Wang, "Iterative phase reconstruction and weighted localization algorithm for indoor RFID-based localization in NLOS environment," *IEEE Sensors J.*, vol. 14, no. 2, pp. 597–611, Feb. 2014.

[25] F. Xiao, Z. Wang, N. Ye, R. Wang, and X.-Y. Li, "One more tag enables fine-grained RFID localization and tracking," *IEEE/ACM Trans. Netw.*, vol. 26, no. 1, pp. 161–174, Feb. 2018.

[26] K. Guo et al., "Ultra-wideband-based localization for quadcopter navigation," *Unmanned Syst.*, vol. 4, no. 1, pp. 23–34, 2016.

[27] A. Chandra et al., "Frequency-domain in-vehicle UWB channel modeling," *IEEE Trans. Veh. Technol.*, vol. 65, no. 6, pp. 3929–3940, Jun. 2016.

[28] S. Shah, K. Chaiwong, L.-O. Kovavisaruch, K. Kaemarungsi, and T. Demeechai, "Antenna delay calibration of UWB nodes," *IEEE Access*, vol. 9, pp. 63294–63305, 2021.

[29] S. Wang, G. Mao, and J. A. Zhang, "Joint time of arrival estimation for coherent UWB ranging in multipath environment with multi user interference," *IEEE Trans. Signal Process.*, vol. 67, no. 14, pp. 3743–3755, Jul. 2019.

[30] J. Tiemann, J. Pillmann, and C. Wietfeld, "Ultra-wideband antenna-induced error prediction using deep learning on channel response data," in *Proc. IEEE 85th Veh. Technol. Conf.*, Sydney, NSW, Australia, 2017, pp. 1–5.

[31] M. Ridolfi, S. Van de Velde, H. Steendam, and E. De Poorter, "Analysis of the scalability of UWB indoor localization solutions for high user densities," *Sensors*, vol. 18, no. 6, p. 1875, 2018.

[32] J.-Y. Lee and R. A. Scholtz, "Ranging in a dense multipath environment using an UWB radio link," *IEEE J. Sel. Areas Commun.*, vol. 20, no. 9, pp. 1677–1683, Dec. 2002.

[33] C. Falsi, D. Dardari, L. Mucchi, and M. Z. Win, "Time of arrival estimation for UWB localizers in realistic environments," *EURASIP J. Appl. Signal Process.*, vol. 2006, no. 1, pp. 1–13, 2006.

[34] L. Yang and G. B. Giannakis, "Ultra-wideband communications: An idea whose time has come," *IEEE Signal Process. Mag.*, vol. 21, no. 6, pp. 26–54, Nov. 2004.

[35] D. Shi, H. Mi, E. G. Collins, and J. Wu, "An indoor low-cost and high-accuracy localization approach for AGVs," *IEEE Access*, vol. 8, pp. 50085–50090, 2020.

[36] K. C. Cheok, M. Radovnikovich, P. Vempaty, G. R. Hudas, J. L. Overholt, and P. Fleck, "UWB tracking of mobile robot," in *Proc. 21st Annu. IEEE Int. Symp. Pers., Indoor Mobile Radio Commun.*, Istanbul, Turkey, 2010, pp. 2615–2620.

[37] Y. Zheng, Q. Zeng, C. Lv, H. Yu, and B. Ou, "Mobile robot integrated navigation algorithm based on template matching VO/IMU/UWB," *IEEE Sensors J.*, vol. 21, no. 24, pp. 27957–27966, Dec. 2021.

[38] J. Li, J. Xue, D. Fu, C. Gui, and X. Wang, "Position estimation and error correction of mobile robots based on UWB and multisensors," *J. Sensors*, vol. 2022, pp. 1–18, Mar. 2022.

[39] O. G. Crespillo, D. Medina, J. Skaloud, and M. Meurer, "Tightly coupled GNSS/INS integration based on robust M-estimators," in *Proc. IEEE/ION Pos. Loc. Nav. Symp.*, Monterey, CA, USA, 2018, pp. 1554–1561.

[40] Q. Fan, B. Sun, Y. Sun, Y. Wu, and X. Zhuang, "Data fusion for indoor mobile robot positioning based on tightly coupled INS/UWB," *J. Navig.*, vol. 70, no. 5, pp. 1079–1097, 2017.

[41] C. Wang, A. Xu, J. Kuang, X. Sui, Y. Hao, and X. Niu, "A high-accuracy indoor localization system and applications based on tightly coupled UWB/INS/floor map integration," *IEEE Sensors J.*, vol. 21, no. 16, pp. 18166–18177, Aug. 2021.

[42] J. Liu, J. Pu, L. Sun, and Z. He, "An approach to robust INS/UWB integrated positioning for autonomous indoor mobile robots," *Sensors*, vol. 19, no. 4, p. 950, 2019.

[43] M.-G. Li, H. Zhu, S.-Z. You, and C.-Q. Tang, "UWB-based localization system aided with inertial sensor for underground coal mine applications," *IEEE Sensors J.*, vol. 20, no. 12, pp. 6652–6669, Jun. 2020.

[44] B. Zhou, H. Fang, and J. Xu, "UWB-IMU-odometer fusion localization scheme: Observability analysis and experiments," *IEEE Sensors J.*, vol. 23, no. 3, pp. 2550–2564, Feb. 2023.

[45] H. Zhang et al., "A dynamic window-based UWB-odometer fusion approach for indoor positioning," *IEEE Sensors J.*, vol. 23, no. 3, pp. 2922–2931, Feb. 2023.

[46] J. González et al., "Mobile robot localization based on ultra-wide-band ranging: A particle filter approach," *Robot. Auton. Syst.*, vol. 57, no. 5, pp. 496–507, 2009.

[47] J. Chaffee and J. Abel, "GDOP and the Cramer-Rao bound," in *Proc. IEEE Pos. Loc. Navig. Symp.*, Las Vegas, NV, USA, 1994, pp. 663–668.

[48] I. Sharp, K. Yu, and Y. J. Guo, "GDOP analysis for positioning system design," *IEEE Trans. Veh. Technol.*, vol. 58, no. 7, pp. 3371–3382, Sep. 2009.

[49] C. Wang, Y. Ning, J. Wang, L. Zhang, J. Wan, and Q. He, "Optimized deployment of anchors based on GDOP minimization for ultra-wideband positioning," *J. Spatial Sci.*, vol. 67, no. 3, pp. 455–472, 2022.

[50] W. Guosheng, Q. Shuqi, L. Qiang, W. Heng, L. Huican, and L. Bing, "UWB and IMU system fusion for indoor navigation," in *Proc. IEEE 37th Chin. Control Conf.*, 2018, pp. 4946–4950.

[51] "The implementation of two-way ranging with the DW1000 version 2.3," Application Note APS013, DecaWave Ltd. Co., Dublin, Ireland, 2015. [Online]. Available: https://www.decawave.com

[52] A. R. Jiménez and F. Seco, "Comparing decawave and bespoon UWB location systems: indoor/outdoor performance analysis," in *Proc. Int. Conf. Indoor Pos. Indoor Navig.*, Alcala de Henares, Spain, 2016, pp. 1–8.

[53] "Antenna delay calibration DW1000-based products and systems version 1.2," Application Note APS014, DecaWave Ltd. Co., Dublin, Ireland, 2019, [Online]. Available: https://www.decawave.com

[54] S. M. Moosavi and S. Ghassabian, "Linearity of calibration curves for analytical methods: A review of criteria for assessment of method reliability," in *Calibration and Validation of Analytical Methods - A Sampling of Current Approaches*. London, U.K.: Intechopen, 2018.

[55] J. Choliz, M. Eguizabal, A. Hernandez-Solana, and A. Valdovinos, "Comparison of algorithms for UWB indoor location and tracking systems," in *Proc. IEEE 73rd Veh. Technol. Conf. (VTC Spring)*, Budapest, Hungary, 2011, pp. 1–5.

[56] Y. Zhou, "An efficient least-squares trilateration algorithm for mobile robot localization," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2009, pp. 3474–3479.

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